

EXERCISE SHEET 1

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

1.1. Let G be a group acting on a set X , and let $x \in X$. Show that the stabiliser

$$G_x := \{g \in G \mid gx = x\},$$

of x in G , is a subgroup.

1.2. Let H be a proper subgroup of a finite group G . Show that G contains at least one element that is not in any conjugate of H .

1.3. Let G be a group of order 77 and let G act on a set X . Show that if $|X| = 60$ and if there is no G -orbit of size 11, then G must fix at least two points in X .

EXTRA QUESTIONS

1.4. Suppose X is a G -set under a given action. Show that $X \times X$ is also a G -set under the action

$$g \cdot (x_1, x_2) = (g \cdot x_1, g \cdot x_2).$$

1.5. Let G be a group with a normal subgroup $N \trianglelefteq G$ and a subgroup $H \leq G$. Suppose H acts on N via a given homomorphism $\varphi : H \rightarrow \text{Aut } N$, where we write $\varphi(h)$ as φ_h for conciseness, and we sometimes also use the notation for conjugation by h , instead of writing φ_h . Then we say that G is the *semidirect product* of N by H , and we write $G \cong N \rtimes_{\varphi} H$, or simply $N \rtimes H$. For $g \in G$ we can write $g = nh$ also as (n, h) , with $n \in N$ and $h \in H$. Then multiplication is written as

$$(n_1, h_1)(n_2, h_2) = (n_1 \cdot \varphi_{h_1}(n_2), h_1 h_2)$$

or

$$(n_1 h_1)(n_2 h_2) = n_1 h_1 n_2 (h_1^{-1} h_1) h_2 = n_1 (h_1 n_2 h_1^{-1}) h_1 h_2 = (n_1 \varphi_{h_1}(n_2)) (h_1 h_2).$$

Determine whether the dihedral groups $D_{2n} = \langle x, y \mid x^n, y^2, yxy = x^{-1} \rangle$, for $n \geq 2$, of order $2n$ are semidirect products. If yes, identify the corresponding subgroup H and normal subgroup N , and determine the action of H on N .

1.6. Prove that in any group the subset of all elements that have only a finite number of conjugates is a subgroup.